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Code ▾

Fix(workflow)/add _process_content_object function in _rehydration_utils file to extract output from event.content object before assigning it to child.output, in _reconstruct_node_states #5909

Closed

[samarth1224](#) wants to merge 1 commit into[google:main](#) from [samarth1224:fix/issue-5553](#) 

Conversation 1



Commits 1



Checks 6



Files changed 2

[samarth1224](#) commented [on May 30](#) • edited ▾

Contributor

Link to Issue

1. Link to an existing issue (if applicable):

- Closes: [Raw event.content assigned to child.output without transformation in _reconstruct_node_states during node re-run, for the nodes with message_as_input = True #5553](#)

NOTE

I opened the pull request before but it was closed due to v2 branch being merged into main branch. The test failures in last PR were due to inherent issues in the v2 branche's unittests.

Problem:

During fresh execution of a node with `message_as_output == True`, `process_llm_agent_output` extracts text from the `Content` parts, parses it, validates it against the schema, and assigns the resulting output to `event.output`.

However, before the event is persisted to the session, `_consume_event_queue` optimizes for `message_as_output` nodes by stripping `event.output` (setting it to `None`) and only saving the raw `event.content`

Upon workflow resumption, `_reconstruct_node_states` rebuilds the node's state. Because `event.output` is `None`, it fell back to assigning the raw `event.content` (a `Content` object) directly to `child.output`.

Solution:

Added `_process_content_object()` to extract the output from the raw `event.content` during rehydration.

This function mirrors the logic used in the `process_llm_agent_output` function in `_llm_agent_wrapper.py`` file

1. It extracts text from all parts of the `Content` object.
2. It correctly filters out `Parts.thought` parts to prevent Chain-of-Thought reasoning text from leaking into the final output.
3. It attempts to parse the extracted text as JSON, if fails it returns the raw text.

Testing Plan

I have added new unit tests for the `_process_content_object` function. These tests cover various scenarios, including plain text extraction, JSON parsing for structured outputs, and the correct filtering of thought parts to prevent internal reasoning from leaking into the final output.

Additionally, I have updated the existing rehydration test `_test_scan_message_as_output` that previously asserted the raw `Content` object was returned; It now correctly verify that the output is reconstructed during node state rehydration.

Unit Tests:

- ✓ I have added or updated unit tests for my change.
- ✓ All unit tests pass locally.

Summary Unit Test

```
tests\unittests\workflow\utils\test_rehydration_utils.py
..... [100%]

===== warnings summary =====
<frozen abc>:106
<frozen abc>:106
<frozen abc>:106
<frozen abc>:106
  <frozen abc>:106: DeprecationWarning: BaseAgentConfig is deprecated and will be removed in
future versions.

-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
===== 40 passed, 4 warnings in 10.20s =====
```



Manual End-to-End (E2E) Tests:

I have already provided the necessary setup and instruction in the description of Issue [#5553](#), along with minimal reproducible code.

The same setup can be used as Manual E2E test.

I am presentig the minimal reprouducible code and successful mitigation of the bug in the screenshot.



```

from google.adk import Workflow
from google.adk.events import RequestInput
from google.adk import Context
from google.adk.agents import Agent
from google.adk.workflow import node, FunctionNode
from pydantic import BaseModel

from typing import Any

class MySchema(BaseModel):
    greeting: str

analyzer = Agent(
    name="my_agent",
    model="gemini-3.1-flash-lite-preview",
    output_schema=MySchema,
    instruction="Your job is to greet the user. "
    "",
)
@node(rerun_on_resume=False)
async def get_user_approval(ctx: Context, node_input: Any):
    """Yields a RequestInput to pause the workflow and wait for user input."""
    yield RequestInput(message=f'please approve or reject.', response_schema = str)

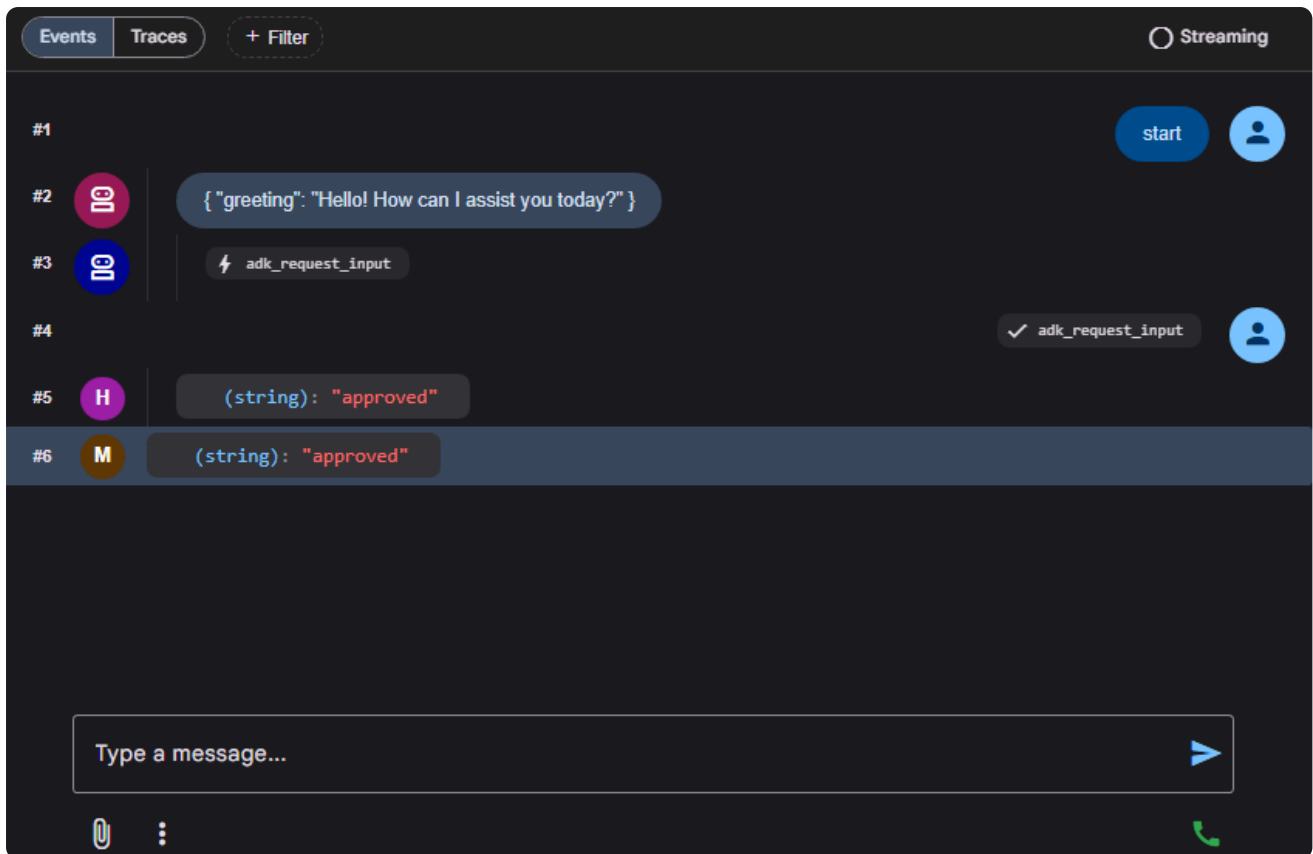
@node(rerun_on_resume=True)
async def handle_process(ctx: Context, node_input: Any):
    """The orchestrator calling the interactive step."""
    user_response = await ctx.run_node(get_user_approval, node_input)
    if user_response.lower() == "yes":
        yield 'approved'
    yield "Denied"
    return

@node(rerun_on_resume=True)
async def my_workflow(ctx: Context):
    agent_input = "my agent input"
    agent_output = await ctx.run_node(analyzer, agent_input)
    agent_output = MySchema.validate(agent_output)
    result = await ctx.run_node(handle_process)
    yield result

root_agent = Workflow(
    name="greet_user",
    edges=[("START", my_workflow)]
)

```

Output



Additional Context

-I have tested it using the the static Graph-Based workflows. The issue remains same, as the raw content is assigned to ctx.output during the rehydration.

-Also as specified in the comments in the `_consume_event_queue` function in `runner.py`

```

async def _consume_event_queue(
    self, ic: InvocationContext, done_sentinel: object
) -> AsyncGenerator[Event, None]:
    """Consume events from ic._event_queue until done_sentinel."""
    while True:
        event_or_done, processed_signal = await ic._event_queue.get()
        if event_or_done is done_sentinel:
            break
        event: Event = event_or_done
        # When an LlmAgent node uses ``message_as_output`` (no
        # ``output_schema``), the wrapper sets both ``event.content``
        # (the model's text) AND ``event.output`` (the same text) to
        # signal that the message IS the node's output. Clear
        # ``event.output`` on a copy here so downstream renderers don't
        # surface the same text twice. Task-mode agents set
        # ``event.output`` from the ``finish_task`` FC args without
        # ``message_as_output``, so this clearing doesn't affect them.
        if not event.partial:
            if event.node_info.message_as_output and event.content is not None:
                event = event.model_copy()
                event.output = None

```



It feels like it intend to set `message_as_output = True` only when the `output_schema` is not provided, but `process_llm_agent_output` function in `llm_agent_wrapper` file, sets the `message_as_output = True` regardless of whether the `output_schema` is provided or not.

Checklist

- ✓ I have read the [CONTRIBUTING.md](#) document.
- ✓ I have performed a self-review of my own code.
- ✓ I have commented my code, particularly in hard-to-understand areas.
- ✓ I have added tests that prove my fix is effective or that my feature works.
- ✓ New and existing unit tests pass locally with my changes.
- ✓ I have manually tested my changes end-to-end.
- ✓ Any dependent changes have been merged and published in downstream modules.



adk-bot added the **core** label on May 30

samarth1224 changed the title [add _process_content_object function in _rehydration_utils file to extract output from event.content object before assigning it to child.output, in _reconstruct_node_states](#) [Fix\(workflow\)add _process_content_object function in _rehydration_utils file to extract output from event.content object before assigning it to child.output, in _reconstruct_node_states](#) on May 30

samarth1224 changed the title [Fix\(workflow\)add _process_content_object function in _rehydration_utils file to extract output from event.content object before assigning it to child.output, in _reconstruct_node_states](#) [Fix\(workflow\)/add _process_content_object function in _rehydration_utils file to extract output from event.content object before assigning it to child.output, in _reconstruct_node_states](#) on May 30

DeanChensj self-assigned this on Jun 3

[add _process_content_object function in _rehydration_utils file to ex...](#) ... ✓ [a01d015](#)

DeanChensj [force-pushed](#) the `fix/issue-5553` branch from `9d2b9a1` to `a01d015` 2 months ago [Compare](#)

copybara-service pushed a commit that referenced this pull request on Jun 3

[fix\(workflow\): Resolve raw Content output crash on rehydration](#) ... ✗ [4f992b0](#)

adk-bot commented [on Jun 3](#)

Collaborator

Thank you [@samarth1224](#) for your contribution! 🎉

Your changes have been successfully imported and merged via Copybara in commit [4f992b0](#) .

Closing this PR as the changes are now in the main branch.



adk-bot added the **merged** label [on Jun 3](#)

adk-bot closed this [on Jun 3](#)

DeanChensj mentioned this pull request [on Jun 3](#)

Raw event.content assigned to child.output without transformation in _reconstruct_node_states during node re-run, for the nodes with message_as_input = True #5553

✓ Closed

samarth1224 deleted the `fix/issue-5553` branch [2 months ago](#)

Restore branch

nagug pushed a commit to `vidaiUK/adk-python` that referenced this pull request [on Jun 6](#)

`fix(workflow): Resolve raw Content output crash on rehydration` ... [98f1ee0](#)

copybara-service Bot pushed a commit that referenced this pull request [on Jun 9](#)

`fix(workflow): Resolve raw Content output crash on rehydration` ... ✗ [428e789](#)

Closed with unmerged commits

This pull request is closed.

Reviewers

No reviews

Assignees

DeanChensj

Labels

core **merged**

Projects

None yet

Milestone

No milestone

Development

Successfully merging this pull request may close these issues.

None yet

3 participants

☐ Allow edits by maintainers 